06 Differentiability

INTRODUCTION OF DERIVABILITY:

Two-fold meaning of derivative

Geometrical meaning of derivative	**Physical meaning of derivative**
Slope of the tangent drawn to the curve At any point (if exists)	Instantaneous rate of change of function

1. MEANING OF DERIVATIVE:

The instantaneous rate of change of a function with respect to the dependent variable is called derivative. Let 'f' be a given function of one variable and let Δx denote a number (positive or negative) to be added to the number x. Let Δf denote the corresponding change of 'f' then $\Delta f = f(x+\Delta x) - f(x)$

$$\Rightarrow \frac{\Delta f}{\Delta x} = \frac{f(x+\Delta x)-f(x)}{\Delta x}$$

If $\Delta f/\Delta x$ approaches a limit as Δx approaches zero, this limit is the derivative of 'f' at the point x. The derivative of a function 'f' is a function; this function is denoted by symbols such as

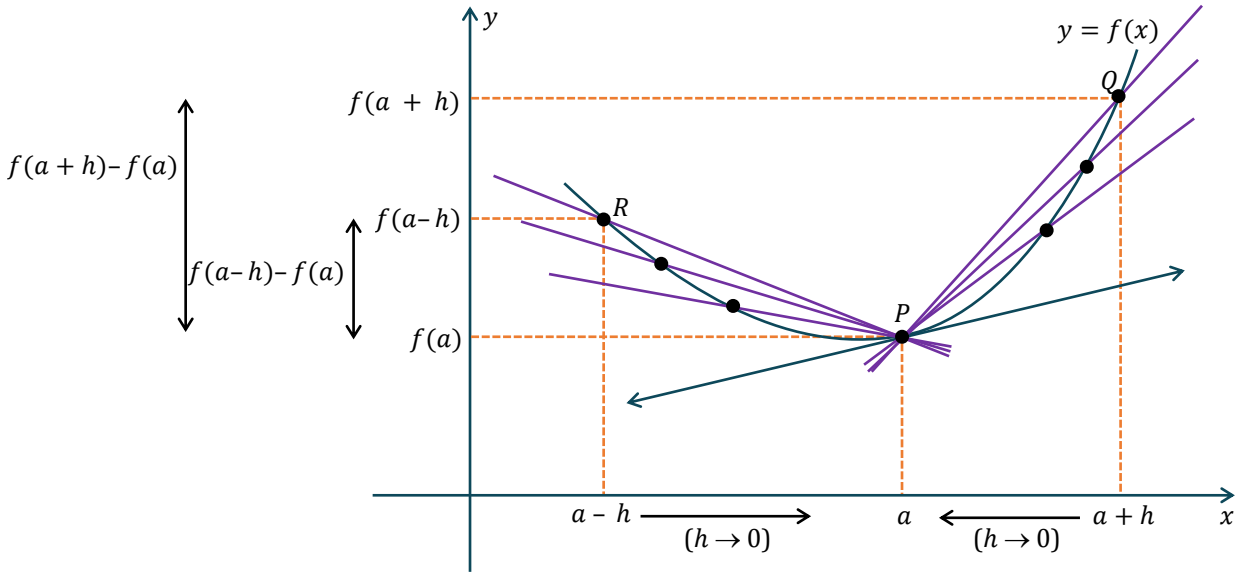
$$f'(x),\ \frac{df}{dx}, \frac{d}{dx}f(x) \text{ or } \frac{df(x)}{dx}$$

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$$\Rightarrow \frac{df}{dx} = \lim_{\Delta x\to 0}\frac{\Delta f}{\Delta x} = \lim_{\Delta x\to 0}\frac{f(x+\Delta x)-f(x)}{\Delta x}$$

The derivative evaluated at a point a, is written, $f'(a), \left.\frac{df(x)}{dx}\right|_{x=a}, f'(x)_{x=a}$, etc.

2. EXISTENCE OF DERIVATIVE AT $x = a$:



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(a) Right hand derivative:

The right hand derivative of $f(x)$ at $x = a$ denoted by $Rf'(a)$ is defined as:

$Rf'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$, provided the limit exists & is finite. $(h > 0)$

(b) Left hand derivative:

The left hand derivative of $f(x)$ at $x = a$ denoted by $Lf'(a)$ is defined as:

$Lf'(a) = \lim_{h\to 0} \frac{f(a-h)-f(a)}{-h}$, provided the limit exists & is finite. $(h > 0)$

Hence $f(x)$ is said to be **derivable or differentiable at $x = a$.**

If $Rf'(a) = Lf'(a)$ = finite quantity and it is denoted by $f'(a)$; where $f'(a) = Lf'(a) = Rf'(a)$ & it is called derivative or differential coefficient of $f(x)$ at $x = a$.

3. DIFFERENTIABILITY & CONTINUITY:

Theorem: If a function $f(x)$ is derivable at $x = a$, then $f(x)$ is continuous at $x = a$.

Proof: $f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$ exists.

Also $f(a+h)-f(a) = \frac{f(a+h)-f(a)}{h}.h \quad [h \neq 0]$

$\therefore \lim_{h\to 0}[f(a+h)-f(a)] = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}.h = f'(a).0 = 0$

$\Rightarrow \lim_{h\to 0}[f(a+h)-f(a)] = 0 \Rightarrow \lim_{h\to 0} f(a+h) = f(a) \Rightarrow f(x)$ is continuous at $x = a$.

Note:

(i) Differentiable $\Rightarrow$ Continuous ; Continuous $\not\Rightarrow$ Differentiable ; Not Differentiable $\not\Rightarrow$ Not Continuous
But Not Continuous $\Rightarrow$ Not Differentiable

(ii) All polynomial, rational, trigonometric, logarithmic and exponential function are continuous and differentiable in their domains.

(iii) If $f(x)$ & $g(x)$ are differentiable at $x = a$, then the functions $f(x) + g(x)$, $f(x) - g(x)$, $f(x).g(x)$ will also be differentiable at $x = a$ & if $g(a) \neq 0$ then the function $f(x)/g(x)$ will also be differentiable at $x = a$.

Illustration 1

Determine the values of x for which the following functions fails to be continuous or differentiable

$$f(x) = \begin{cases} (1-x), & x < 1 \\ (1-x)(2-x), & 1 \le x \le 2 \\ (3-x), & x > 2 \end{cases}$$. Justify your answer.

Solution:

By the given definition it is clear that the function f is continuous and differentiable at all points except possibility at $x = 1$ and $x = 2$.

Check the differentiability at $x = 1$

$\Rightarrow q = \text{LHD} = \lim_{h\to 0} \frac{f(1-h)-f(1)}{-h} = \lim_{h\to 0} \frac{1-(1-h)-0}{-h} = -1$

$\Rightarrow p = \text{RHD} = \lim_{h\to 0} \frac{f(1+h)-f(1)}{h} = \lim_{h\to 0} \frac{\{1-(1+h)\}\{2-(1+h)\}-0}{h} = -1$

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$\because q = p \ \therefore$ Differentiable at $x = 1 \Rightarrow$ Continuous at $x = 1$.

Check the differentiability at $x = 2$

$\Rightarrow q$ = LHD = $\lim\limits_{h\to 0}\dfrac{f(2-h)-f(2)}{-h}=\lim\limits_{h\to 0}\dfrac{(1-2+h)(2-2+h)-0}{-h}=1$ = finite

$\Rightarrow p$ = RHD = $\lim\limits_{h\to 0}\dfrac{f(2+h)-f(2)}{h}=\lim\limits_{h\to 0}\dfrac{(3-2-h)-0}{h}\to\infty$ (not finite)

$\because q \neq p \ \therefore$ not differentiable at $x = 2$.

Now we have to check the continuity at $x = 2$

$\Rightarrow$ LHL = $\lim\limits_{x\to 2^-} f(x)=\lim\limits_{x\to 2^-}(1-x)(2-x)=\lim\limits_{h\to 0}(1-(2-h))(2-(2-h))=0$

$\Rightarrow$ RHL = $\lim\limits_{x\to 2^+} f(x)=\lim\limits_{x\to 2^+}(3-x)=\lim\limits_{h\to 0}(3-(2+h))=1$

$\therefore$ LHL $\neq$ RHL

$\Rightarrow$ not continuous at $x = 2$.

Illustration 2

$f(x)=\big||x|-1\big|$ is not differentiable at

(A) 0 (B) ±1,0 (C) 1 (D) ±1

Ans. (B)

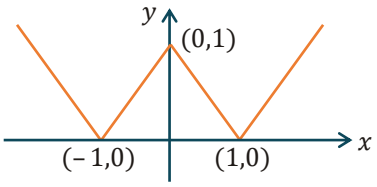
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Solution:

$$=\begin{cases}|x|-1, & |x|-1\geq 0\\ -|x|+1, & |x|-1<0\end{cases}$$

$$=\begin{cases}|x|-1, & x\leq -1 \text{ or } x\geq 1\\ -|x|+1, & -1<x<1\end{cases}$$

$$=\begin{cases}-x-1, & x\leq -1\\ x+1, & -1<x<0\\ -x+1, & 0\leq x<1\\ x-1, & x\geq 1\end{cases}$$



From the graph. It is clear that $f(x)$ is not differentiable at $x = -1,0$ and 1.

Illustration 3

For what triplets of real numbers (a, b, c) with $a \neq 0$ the function

$f(x)=\begin{cases}x, & x\leq 1\\ ax^2+bx+c, & \text{otherwise}\end{cases}$ is differentiable for all real x?

(A) $\{(a, 1-2a, a)|a \in R, a \neq 0\}$ (B) $\{(a, 1-2a, c)|a, c \in R, a \neq 0\}$

(C) $\{(a, b, c)|a, b, c \in R, a+b+c = 1\}$ (D) $\{(a, 1-2a, 0)|a \in R, a \neq 0\}$

Ans. (A)

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Solution:

$$f(x)=\begin{cases} x & , \quad x\le 1 \\ ax^2+bx+c & , \quad \text{otherwise} \end{cases}$$

$\Rightarrow f(x)$ should be continuous at $x = 1$

it gives $a + b + c = 1$

$\Rightarrow f(x)$ should be differentiable at $x = 1$

it gives $2a + b = 1 \Rightarrow b = 1 - 2a$

$\Rightarrow c = 1 - a - b = a$

Illustration 4

Test the continuity & differentiability of the function defined as under at x = 1 & x = 2.

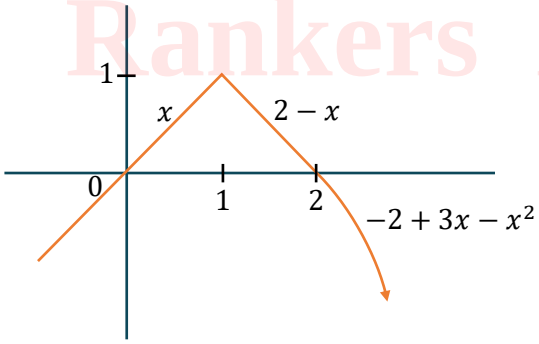
$$f(x) = \begin{cases} x & ; \quad x < 1 \\ 2 - x & ; \quad 1 \le x \le 2 \\ -2 + 3x - x^2 & ; \quad x > 2 \end{cases}$$

Ans. (Continuous at both points but differentiable only at x = 2)

Solution:

Curve of $y = f(x)$ is as follows

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It is continuous every where not differentiable at x = 1

LHD (x = 2) = – 1 = RHD (x = 2) diff. at x = 2

Illustration 5

A function f is defined as follows: $f(x) = \begin{cases} 1 & \text{for} \quad -\infty < x < 0 \\ 1 + \sin x & \text{for} \quad 0 \le x < \dfrac{\pi}{2} \\ 2 + \left(x - \dfrac{\pi}{2}\right)^2 & \text{for} \quad \dfrac{\pi}{2} \le x < \infty \end{cases}$

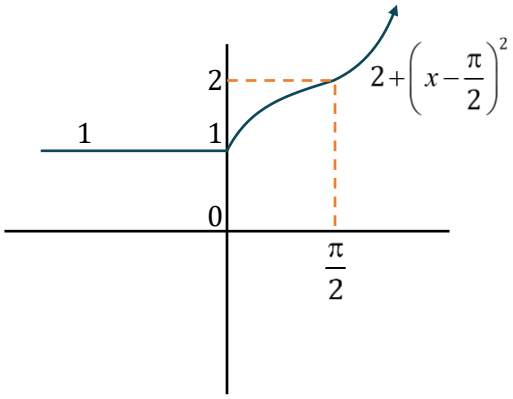
Discuss the continuity & differentiability at x = 0 & $x = \pi/2$.

Ans. (continuous but not differentiable at x = 0; differentiable & continuous at $x = \pi/2$)

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Solution:

$y = f(x)$ is as follows



continuous every where

non-differentiable at $x = 0$

Illustration 6

Prove that $f(x) = |x| \cos x$ is not differentiable at $x = 0$

Solution:

$$f(x) = |x| \cos x = \begin{bmatrix} x \cos x & , & x \geq 0 \\ -x\cos x & , & x < 0 \end{bmatrix}$$

$f(x)$ is non differentiable at $x = 0$

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Illustration 7

Examine the differentiability of $f(x) = \sqrt{1-e^{-x^2}}$ at $x = 0$.

Ans. (not differentiable at $x = 0$)

Solution:

$f(x) = \sqrt{1-e^{-x^2}}$ $\qquad$ $f(0) = 0$

$$\text{LHD } (x=0) = \lim_{h\to 0} \frac{\sqrt{1-e^{-h^2}}-0}{-h} = \lim_{h\to 0} -\sqrt{1+\frac{h^2}{2\ !}+\ldots} = -1$$

$$\text{RHD } (x=0) = \lim_{h\to 0} \frac{\sqrt{1-e^{-h^2}}-0}{h} = \lim_{h\to 0} \sqrt{1+\frac{h^2}{2\ !}+\ldots} = 1$$

Not differentiable at $x = 0$

Important Note:

(a) Let $Rf'(a) = p$ & $Lf'(a) = q$ where p & q are finite then:

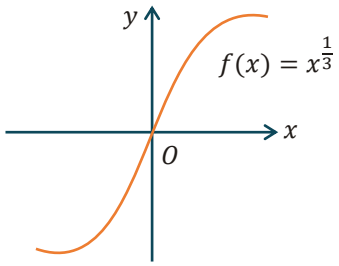
(i) $p = q \Rightarrow f$ is differentiable at $x = a \Rightarrow f$ is continuous at $x = a$

(ii) $p \neq q \Rightarrow f$ is not differentiable at $x = a$, but f is continuous at $x = a$.

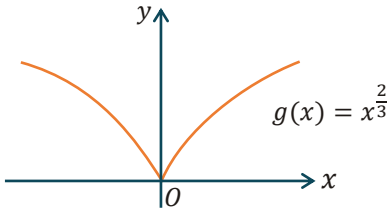
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(b) Vertical Tangent:

(i) If $y = f(x)$ is continuous at $x = a$ and $\lim_{x \to a} |f'(x)|$ approaches to ∞, then $y = f(x)$ has a vertical tangent at $x = a$. If a function has vertical tangent at $x = a$ then it is non differentiable at $x = a$.



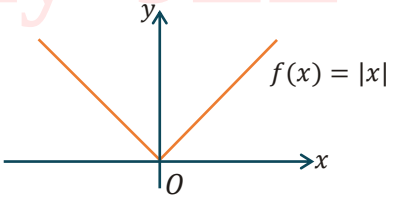
e.g. (1) $f(x) = x^{\frac{1}{3}}$ has vertical tangent at $x = 0$ since $f_+'(0) \to \infty$ and $f_-'(0) \to \infty$ hence $f(x)$ is not differentiable at $x = 0$



(2) $g(x) = x^{\frac{2}{3}}$ have vertical tangent at $x = 0$ since $g_+'(0) \to \infty$ and $g_-'(0) \to -\infty$ hence $g(x)$ is not differentiable at $x = 0$.

(c) Geometrical Interpretation of Differentiability:

(i) If the function $y = f(x)$ is differentiable at $x = a$, then a unique non vertical tangent can be drawn to the curve $y = f(x)$ at the point $P(a, f(a))$ & $f'(a)$ represent the slope of the tangent at point P.



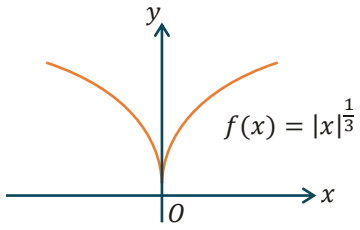
(ii) If a function $f(x)$ does not have a unique tangent (p & q are finite but unequal), then f is continuous at $x = a$, it geometrically implies a corner at $x = a$.

e.g. $f(x) = |x|$ is continuous but not differentiable at $x = 0$ & there is corner at $x = 0$.

(does not have unique tangent, corner at $x=0$) $\begin{cases} p = 1 \\ q = -1 \end{cases}$

(iii) If one of p & q tends to ∞ and other tends to $-\infty$, then their will be a cusp at $x = a$. Where $p = Rf'(a)$ and $q = Lf'(a)$

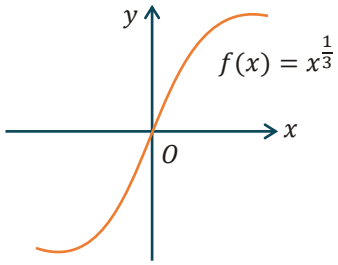
e.g. (1) $f(x) = |x|^{\frac{1}{3}}$ is continuous but not differentiable at $x = 0$ & there is cusp at $x = 0$.



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(has a vertical tangent, cusp at $x = 0$) $\begin{cases} p \to -\infty \\ q \to +\infty \end{cases}$

(2) $f(x) = x^{\frac{1}{3}}$ is continuous but not differentiable at $x = 0$ because $Rf'(0) \to \infty$ and $Lf'(0) \to \infty$.



(have a unique vertical tangent but does not have corner) $\begin{cases} p \to +\infty \\ q \to +\infty \end{cases}$

Note: corner/vertical tangent $\Rightarrow$ non differentiable

non differentiable $\not\Rightarrow$ corner/vertical tangent

Illustration 8

$f(x) = \begin{cases} [\cos \pi x] & x \le 1 \\ 2\{x\} - 1 & x > 1 \end{cases}$ comment on the derivability at $x = 1$, where [.] denotes greatest integer function & {.} denotes fractional part function.

Solution:

$$\Rightarrow Lf'(1) = \lim_{h \to 0} \frac{f(1-h) - f(1)}{-h} = \lim_{h \to 0} \frac{[\cos(\pi - \pi h)] + 1}{-h} = \lim_{h \to 0} \frac{-1+1}{-h} = 0$$

$$\Rightarrow Rf'(1) = \lim_{h \to 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \to 0} \frac{2\{1+h\} - 1 + 1}{h} = \lim_{h \to 0} \frac{2h}{h} = 2$$

Hence $f(x)$ is not differentiable at $x = 1$.

Illustration 9

Let $f : (-1,1) \to R$ be a function defined by $f(x) = \max\left\{-|x|, -\sqrt{1-x^2}\right\}$. If K be the set of all points at which f is not differentiable, then K has exactly :

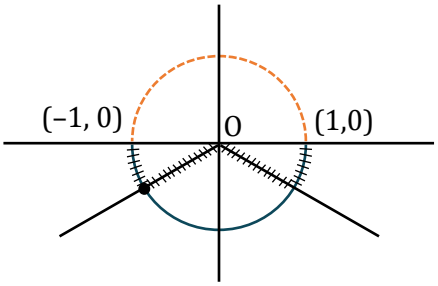
(A) Three elements (B) One element (C) Five elements (D) Two elements

Ans. (A)

Solution:

$f : (-1, 1) \to R$

$f(x) = \max\left\{-|x|, -\sqrt{1-x^2}\right\}$



Non-derivable at 3 points in (–1, 1)

Option (1)

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Illustration 10

Let S be the set of all points in $(-\pi,\pi)$ at which the function, $f(x) = \min\{\sin x, \cos x\}$ is not differentiable. Then S is a subset of which of the following?

(A) $\left\{-\frac{3\pi}{4},-\frac{\pi}{4},\frac{3\pi}{4},\frac{\pi}{4}\right\}$ (B) $\left\{-\frac{3\pi}{4},-\frac{\pi}{2},\frac{\pi}{2},\frac{3\pi}{4}\right\}$ (C) $\left\{-\frac{\pi}{2},-\frac{\pi}{4},\frac{\pi}{4},\frac{\pi}{2}\right\}$ (D) $\left\{-\frac{\pi}{4},0,\frac{\pi}{4}\right\}$

Ans. (A)

Solution:

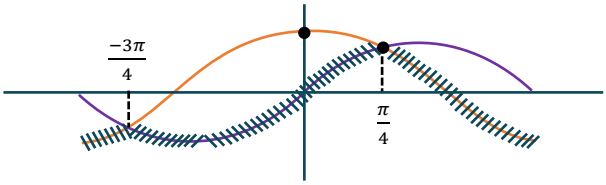


Illustration 11

Let $[t]$ denote the greatest integer less than or equal to t. Let $f(x) = x - [x]$, $g(x) = 1 - x + [x]$, and $h(x) = \min\{f(x), g(x)\}$, $x \in [-2, 2]$. Then h is :

(A) continuous in $[-2, 2]$ but not differentiable at more than four points in $(-2, 2)$

(B) not continuous at exactly three points in $[-2, 2]$

(C) continuous in $[-2, 2]$ but not differentiable at exactly three points in $(-2, 2)$

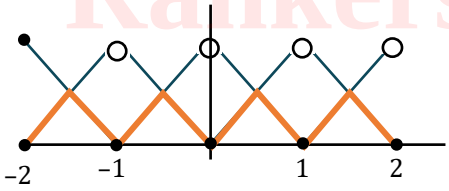
(D) not continuous at exactly four points in $[-2, 2]$

Ans. (A)

Solution:

$\min\{x - [x], 1 - x + [x]\}$

$h(x) = \min\{x - [x], 1 - [x - [x]]\}$



$\Rightarrow$ always continuous in $[-2, 2]$

but non differentiable at 7 Points

Illustration 12

If $f(x) = \begin{cases} x-3 & x<0 \\ x^2-3x+2 & x\geq 0 \end{cases}$. Draw the graph of the function & discuss the continuity and differentiability of $f(|x|)$ and $|f(x)|$.

Solution:

$$f(|x|) = \begin{cases} |x|-3; & |x|<0 \rightarrow \text{not possible} \\ |x|^2-3|x|+2; & |x|\geq 0 \end{cases}$$

$$\Rightarrow f(|x|) = \begin{cases} x^2+3x+2, & x<0 \\ x^2-3x+2, & x\geq 0 \end{cases}$$

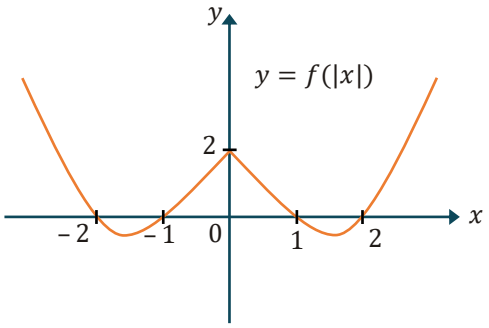
at $x = 0$

$$\Rightarrow q=\text{LHD}=\lim_{h\to 0}\frac{f(0-h)-f(0)}{-h}=\lim_{h\to 0}\frac{h^2-3h+2-2}{-h}=3$$

$$\Rightarrow p=\text{RHD}=\lim_{h\to 0}\frac{f(0+h)-f(0)}{h}=\lim_{h\to 0}\frac{h^2-3h+2-2}{h}=-3$$

$\Rightarrow q \neq p$

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$\therefore$ not differentiable at $x = 0$. but p & q are both are finite

$\Rightarrow$ continuous at $x = 0$

$$\text{Now, } |f(x)| = \begin{cases} 3 - x, & x < 0 \\ (x^2 - 3x + 2), & 0 \le x < 1 \\ -(x^2 - 3x + 2), & 1 \le x \le 2 \\ (x^2 - 3x + 2), & x > 2 \end{cases}$$

To check differentiability at $x = 0$

$$\left.\begin{aligned} q = \text{LHD} &= \lim_{h\to 0} \frac{f(0-h) - f(0)}{-h} = \lim_{h\to 0} \frac{3 + h - 2}{-h} = \lim_{h\to 0} \frac{(1+h)}{-h} \to -\infty \\ p = \text{RHD} &= \lim_{h\to 0} \frac{f(0+h) - f(0)}{h} = \lim_{h\to 0} \frac{h^2 - 3h + 2 - 2}{h} = -3 \end{aligned}\right\} \Rightarrow \text{not differentiable at } x = 0$$

Now to check continuity at $x = 0$

$$\left.\begin{aligned} \text{LHL} &= \lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} 3 - x = 3 \\ \text{RHL} &= \lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} x^2 - 3x + 2 = 2 \end{aligned}\right\} \Rightarrow \text{not continuous at } x = 0$$

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To check differentiability at $x = 1$

$$\Rightarrow q = \text{LHD} = \lim_{h\to 0} \frac{f(1-h) - f(1)}{-h}$$

$$\Rightarrow \lim_{h\to 0} \frac{(1-h)^2 - 3(1-h) + 2 - 0}{-h} = \lim_{h\to 0} \frac{h^2 + h}{-h} = -1$$

$$\Rightarrow p = \text{RHD} = \lim_{h\to 0} \frac{f(1+h) - f(1)}{h} = \lim_{h\to 0} \frac{-(h^2 + 2h + 1 - 3 + 3h + 2) - 0}{h} = \lim_{h\to 0} \frac{-(h^2 - h)}{h} = 1$$

$\Rightarrow$ not differentiable at $x = 1$.

but $|f(x)|$ is continuous at $x = 1$, because $p \neq q$ and both are finite.

To check differentiability at $x = 2$

$$\Rightarrow q = \text{LHD} = \lim_{h\to 0} \frac{f(2-h) - f(2)}{-h}$$

$$\Rightarrow \lim_{h\to 0} \frac{-(4 + h^2 - 4h - 6 + 3h + 2) - 0}{-h} = \lim_{h\to 0} \frac{h^2 - h}{h} = -1$$

$$\Rightarrow p = \text{RHD} = \lim_{h\to 0} \frac{f(2+h) - f(2)}{h} = \lim_{h\to 0} \frac{(h^2 + 4h + 4 - 6 - 3h + 2) - 0}{h} = \lim_{h\to 0} \frac{(h^2 + h)}{h} = 1$$

$\Rightarrow$ not differentiable at $x = 2$.

but $|f(x)|$ is continuous at $x = 2$, because $p \neq q$ and both are finite.

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4. DERIVABILITY OVER AN INTERVAL:

$f(x)$ is said to be differentiable over an open interval (a, b) if it is differentiable at each & every point of the open interval (a, b)

$f(x)$ is said to be differentiable over the closed interval $[a, b]$ if

(i) $f(x)$ is differentiable in (a, b) &

(ii) for the points a and b, $f'_+(a)$ & $f'_-(a)$ exist.

Note:

(i) If $f(x)$ is differentiable at $x = a$ & $g(x)$ is not differentiable at $x = a$, then the product function $F(x) = f(x).g(x)$ can still be differentiable at $x = a$.
e.g. Consider $f(x) = x$ & $g(x) = |x|$. f is differentiable at $x = 0$ & g is non-differentiable at $x = 0$, but $f(x).g(x)$ is still differentiable at $x = 0$.

(ii) If $f(x)$ & $g(x)$ both are not differentiable at $x = a$ then the product function; $F(x) = f(x).g(x)$ can still be differentiable at $x = a$.
e.g. Consider $f(x) = |x|$ & $g(x) = -|x|$. f & g are both non-differentiable at $x = 0$, but $f(x).g(x)$ still differentiable at $x = 0$.

(iii) If $f(x)$ & $g(x)$ both are non-differentiable at $x = a$ then the sum function $F(x) = f(x) + g(x)$ may be a differentiable function.
e.g. $f(x) = |x|$ & $g(x) = -|x|$. f & g are both non-differentiable at $x = 0$, but $(f + g)(x)$ still differentiable at $x = 0$.

(iv) If $f(x)$ is differentiable at $x = a \Rightarrow f'(x)$ is continuous at $x = a$.

e.g. $f(x) = \begin{cases} x^2 \sin\frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

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Illustration 13

If $f(x) = \begin{cases} e^{-|x|}, & -5 < x < 0 \\ -e^{-|x-1|} + e^{-1} + 1, & 0 \le x < 2 \\ e^{-|x-2|}, & 2 \le x < 4 \end{cases}$. Discuss the continuity and differentiability of $f(x)$ in the interval $(-5, 4)$.

Solution:

$$f(x) = \begin{cases} e^{+x} & -5 < x < 0 \\ -e^{x-1} + e^{-1} + 1 & 0 \le x \le 1 \\ -e^{-x+1} + e^{-1} + 1 & 1 < x < 2 \\ e^{-x+2} & 2 \le x < 4 \end{cases}$$

Check the differentiability at $x = 0$

$$\Rightarrow \text{LHD} = \lim_{h\to 0} \frac{f(0-h) - f(0)}{-h} = \lim_{h\to 0} \frac{e^{-h} - 1}{-h} = 1$$

$$\Rightarrow \text{RHD} = \lim_{h\to 0} \frac{f(0+h) - f(0)}{h} = \lim_{h\to 0} \frac{-e^{h-1} + e^{-1} + 1 - 1}{h} = -e^{-1}$$

$\Rightarrow$ LHD $\neq$ RHD

$\therefore$ Not differentiable at $x = 0$, but continuous at $x = 0$ since LHD and RHD both are finite.

Check the differentiability at $x = 1$

$$\Rightarrow \text{LHD} = \lim_{h\to 0} \frac{f(1-h) - f(1)}{-h} = \lim_{h\to 0} \frac{-e^{1-h-1} + e^{-1} + 1 - e^{-1}}{-h} = -1$$

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$\Rightarrow$ RHD $= \lim\limits_{h\to 0}\dfrac{f(1+h)-f(1)}{h} = \lim\limits_{h\to 0}\dfrac{-e^{1-h-1}+e^{-1}+1-e^{-1}}{h} = 1$

$\Rightarrow$ LHD $\neq$ RHD

$\therefore$ Not differentiable at $x = 1$, but continuous at $x = 1$ since LHD and RHD both are finite.

Check the differentiability at $x = 2$

$\Rightarrow$ LHD$=\lim\limits_{h\to 0}\dfrac{f(2-h)-f(2)}{-h} = \lim\limits_{h\to 0}\dfrac{-e^{-2+h+1}+e^{-1}+1-1}{-h} = \lim\limits_{h\to 0}\dfrac{-e^{-1}(e^{h}-1)}{-h} = e^{-1}$

$\Rightarrow$ RHD$=\lim\limits_{h\to 0}\dfrac{f(2+h)-f(2)}{h} = \lim\limits_{h\to 0}\dfrac{e^{-h}-1}{h} = -1$

$\Rightarrow$ LHD $\neq$ RHD

$\therefore$ Not differentiable at $x = 2$, but continuous at $x = 2$ since LHD & RHD both are finite.

Illustration 14

Number of points of non-differentiability of the function

$g(x) = [x^2]\{\cos^2 4x\} + \{x^2\}[\cos^2 4x] + x^2\sin^2 4x + [x^2][\cos^2 4x] + \{x^2\}\{\cos^2 4x\}$

In $(-50, 50)$ where $[x]$ and $\{x\}$ denotes the greatest integer function and fractional part function of x respectively, is equal to :-

(A) 98 (B) 99 (C) 100 (D) 0

Ans. (D)

Solution:

$g(x) = \{\cos^2 4x\}([x^2] + \{x^2\}) + [\cos^2 4x]([x^2] + \{x^2\}) + x^2\sin^2 4x$

(Using $x = [x] + \{x\}$)

$\Rightarrow g(x) = x^2\{\cos^2 4x\} + x^2[\cos^2 4x] + x^2\sin^2 4x$

$\Rightarrow g(x) = x^2(\{\cos^2 4x\} + [\cos^2 4x]) + x^2\sin^2 4x$

$\Rightarrow g(x) = x^2(\cos^2 4x + \sin^2 4x)$

$\Rightarrow g(x) = x^2$

$g(x)$ is always differentiable.

Illustration 15

The function $f(x) = (x^2-1)|x^2-3x+2| + \cos(|x|)$ is NOT differentiable at :

(A) –1 (B) 0 (C) 1 (D) 2

Ans. (D)

Solution:

$f(x) = (x^2-1)|(x-1)(x-2)|$

Clearly not differentiable at $x = 2$

Illustration 16

Let S denotes the set of all points where $\sqrt[5]{x^2|x|^3} - \sqrt[3]{x^2|x|} - 1$ is not differentiable then S is a subset of –

(A) $\{0, 1\}$ (B) $\{0, 1, -1\}$ (C) $\{0, 1\}$ (D) $\{0\}$

Ans. (A,B,C,D)

Solution:

$f(x) = \sqrt[5]{|x|^5} - \sqrt[3]{|x|^3} - 1$

$= |x| - |x| - 1$

$f(x)$ is always differentiable $\forall\ x \in \mathbb{R}$

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Illustration 17

Draw a graph of the function, $y = [x] + |1 - x|$, $-1 \le x \le 3$. Determine the points, if any, where this function is not differentiable, where $[\cdot]$ denotes the greatest integer function.

Ans. (f is not derivable at all integral values in $-1 < x \le 3$)

Solution:

$y = [x] + |1 - x|$

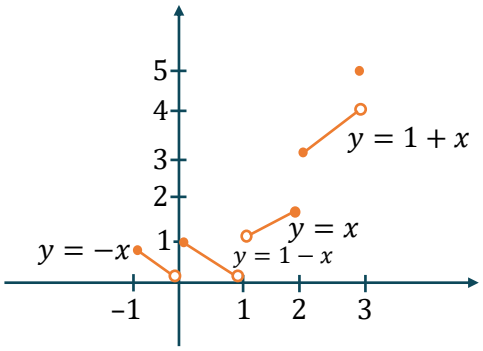


Illustration 18

Discuss the continuity & derivability of $f(x) = \begin{cases} \left| x - \frac{1}{2} \right| & ; \ 0 \le x < 1 \\ x \ . \ [x] & ; \ 1 \le x \le 2 \end{cases}$

where $[x]$ indicates the greatest integer x.

Ans. (f is continuous but not derivable at $x = 1/2$, f is neither differentiable nor continuous at $x = 1$ & $x = 2$)

Solution:

$y = f(x)$

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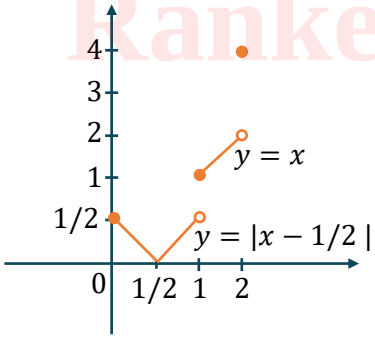


Illustration 19

Discuss continuity and differentiability of $y = f(x)$ in $[-2, 5]$

where [.] denotes GIF & {.} denotes FPF $f(x) = \begin{cases} [x] & , \ x \in [-2, \ 0] \\ \{x\} & , \ x \in (0, \ 2) \\ \frac{x^2}{4} & , \ x \in [2, \ 3) \\ \frac{1}{\log_4(x-3)} & , \ x \in [3, \ 5] \end{cases}$

Ans. (discontinuous and non-differentiable at $-1, 0, 1, 3, 4$)

Solution:

$[x]$ is dicontinuous at $-1, 0$

$\{x\}$ is dicontinuous at 1

$\frac{1}{\log_4(x-3)}$ is discontinuous at 3, 4

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Illustration 20

Check differentiability of $f(x) = \text{sgn}(x^{2/3}) + \left[\cos\left(\frac{x^2}{1+x^2}\right)\right] + |x-1|^{5/3}$ in $[-2, 2]$ where [.] denotes GIF.

Ans. (Differentiable in $[-2, 2]$**)**

Solution:

$|x-1|^{5/3}$ is differentiable for all x

Let $g(x) = \text{sgn}(x^{2/3}) + \left[\cos\left(\frac{x^2}{1+x^2}\right)\right]$

$$\Rightarrow g'(0) = \lim_{h\to 0} \frac{\text{sgn}\left(h^{2/3}\right)+\left[\cos\left(\frac{h^2}{1+h^2}\right)\right]-1}{h}$$

$$= \lim_{h\to 0} \frac{1+0-1}{h} = 0$$

So differentiable at 0.

Illustration 21

Discuss the continuity and differentiability of $h(x) = f(x)\, g(x)$ in $(0, 3)$ if

$$f(x) = \frac{e^x - e}{[x]+1} \text{ {where [.] denot GIF} and } g(x) = \begin{cases} \frac{|x-1|+|x-2|}{2} & , \quad x\in(0,1) \\ |x-1|+|x-2| & , \quad x\in[1,2) \\ \frac{3(|x-1|+|x-2|)}{2} & , \quad x\in[2,3) \end{cases}$$

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Ans. (Continues everywhere in $(0, 3)$ but non differentiable at $x = 2$**)**

Solution:

$$h(x) = f(x)\, g(x) = \begin{cases} \frac{(e^x - e)(3-2x)}{2} & , \quad x\in(0,1) \\ \frac{(e^x - e)}{2} & , \quad x\in[1,2) \\ \frac{(e^x - e)(2x-3)}{2} & , \quad x\in[2,3) \end{cases}$$

$$h'(x) = \begin{cases} \frac{e^x(3-2x)-2(e^x-e)}{2} & , \quad x\in(0,1) \\ e/2 & , \quad x=1 \\ \frac{e^x}{2} & , \quad x\in(1,2) \\ \text{Not Defined} & , \quad x=2 \\ \frac{e^x(2x-3)+2(e^x-e)}{2} & , \quad x\in(2,3) \end{cases}$$

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5. IMPORTANT CONCEPTS:

Let $f(x)$ be a function whose derivative is $f'(x)$ then $\lim\limits_{y\to x}\frac{f(x)-f(y)}{x-y}=f'(x)$

6. DETERMINATION OF FUNCTION SATISFYING THE GIVEN FUNCTIONAL EQUATION:

Write down the expression for $f'(x)$ as $f'(x)=\frac{f(x+h)-f(x)}{h}$

Manipulate $f(x+h)-f(x)$ in such a way that the given functional rule is applicable. Now apply the functional rule and simplify the R.H.S to get $f'(x)$ as a function of x along with constant if any Integrate $f'(x)$ to get $f(x)$ as a function of x and a constant of integration. In some cases a Differential Equation is formed which can be solved to get $f(x)$ Apply the boundary value conditions to determine the value of the constant

Illustration 22

A derivable function $f:\mathbb{R}^+\to\mathbb{R}$ satisfies the condition $f(x)-f(y)\geq \ell n\left(\frac{x}{y}\right)+x-y$ for every $x,y\in\mathbb{R}^+$.

If g denotes the derivative of f then compute the value of the sum $\sum\limits_{n=1}^{100}g\left(\frac{1}{n}\right)$.

Solution:

Let $x=y+h$, then

$f(y+h)-f(y)\geq \ell n(y+h)-\ell ny+h$

$\Rightarrow \lim\limits_{h\to0}\frac{f(y+h)-f(y)}{h}\geq\lim\limits_{h\to0}\frac{\ell n(y+h)-\ell ny}{h}+1$

$\Rightarrow f'(y)\geq\frac{d}{dx}(\ell ny)+1$ ...(i)

Put $y=x+h$ then

$\Rightarrow f(x)-f(x+h)\geq \ell n(x)-\ell n(x+h)-h$

$\Rightarrow f(x+h)-f(x)\leq \ell n(x+h)-\ell nx+h$

$\Rightarrow \lim\limits_{h\to0}\frac{f(x+h)-f(x)}{h}\leq\lim\limits_{h\to0}\frac{\ell n(x+h)-\ell nx}{h}+1$

$\Rightarrow f'(x)\leq\frac{d}{dx}(\ell nx)+1=\frac{1}{x}+1$ ...(ii)

From (i) & (ii)

$\Rightarrow f'(x)=\frac{1}{x}+1=g(x)\Rightarrow g\left(\frac{1}{n}\right)=n+1$

$\Rightarrow \sum\limits_{n=1}^{100}g\left(\frac{1}{n}\right)=\sum\limits_{n=1}^{100}(n+1)$

$\Rightarrow\frac{100\times101}{2}+100$

$\Rightarrow 5050+100$

$\Rightarrow 5150$

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Illustration 23

If $\lim_{x\to 0}\frac{f(3-\sin x)-f(3+x)}{x}=8$, then $|f'(3)|$ is

Solution:

$$\lim_{x\to 0}\frac{f(3-\sin x)-f(3)-(f(3+x)-f(3))}{x}$$

$\Rightarrow -f'(3)-f'(3)=8 \Rightarrow f'(3)=-4$

$\therefore |f'(3)|=4$

Illustration 24

Let f be differentiable at $x=0$ and $f'(0)=1$. Then $\lim_{h\to 0}\frac{f(h)-f(-2h)}{h}=$

(A) 3 (B) 2 (C) 1 (D) –1

Ans. (A)

Solution:

$$\lim_{h\to 0}\frac{f(h)-f(-2h)}{h}$$

$$\Rightarrow \lim_{h\to 0}\frac{f(0+h)-f(0)}{h}+\frac{2(f(0-2h)-f(0))}{(-2h)}$$

$\Rightarrow f'(0)+2f'(0)=3f'(0)=3$

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Illustration 25

If $2x+3|y|=4y$, then y as a function of x i.e. $y=f(x)$, is -

(A) discontinuous at one point

(B) non-differentiable at one point

(C) discontinuous & non-differentiable at same point

(D) continuous & differentiable everywhere

Ans. (B)

Solution:

(I) Let $y\geq 0$

$2x+3y=4y$

$y=2x \Rightarrow x\geq 0$

(II) Let $y<0$

$2x-3y=4y$

$7y=2x$

$y=\frac{2x}{7}\Rightarrow x<0$

$$\Rightarrow y=\begin{cases}2x; & x\geq 0\\ \frac{2x}{7}; & x<0\end{cases}$$

$\Rightarrow$ Non-differentiable at $x=0$

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Illustration 26

Let $\mathbb{R}$ be the set of real numbers and $f:\mathbb{R}\to\mathbb{R}$, be a differentiable function such that $|f(x)-f(y)|\le|x-y|^3\ \forall\ x,y\in\mathbb{R}$. If $f(10)\ =100$, then the value of $f(20)$ is equal to -

(A) 0 (B) 10 (C) 20 (D) 100

Ans. (D)

Solution:

Replace x by $x+h$ & y by x where $h\to0$

$\Rightarrow\lim\limits_{h\to0}|f(x+h)-f(x)|\le|h|^3$

$\Rightarrow\lim\limits_{h\to0}\left|\dfrac{f(x+h)-f(x)}{h}\right|\le|h^2|$

$\Rightarrow|f'(x)|\le0\Rightarrow f'(x)=0;f(x)=$ constant

$\Rightarrow f(x)=100$

So $f(20)=100$

Illustration 27

Let $f(x)$ be a differentiable function such that $2f(x+y)+f(x-y)=3f(x)+3f(y)+2xy\ \forall\ x,y\in\mathbb{R}$ & $f'(0)=0$, then $f(10)+f'(10)$ is equal to

Solution:

$2f(x+y)+f(x-y)=3f(x)+3f(y)+2xy$

Diff. w.r.t 'y' we get

$\Rightarrow 2f'(x+y)-f'(x-y)=3f'(y)+2x$

Also $f(0)=0$ So, put $y=0$.

$\Rightarrow 2f'(x)-f'(x)=3f'(0)+2x$

$\Rightarrow f'(x)=2x\Rightarrow f(x)=x^2+\lambda$

$\Rightarrow f(0)=0\Rightarrow\lambda=0$

$\Rightarrow f(x)=x^2$

$\Rightarrow f(10)=100\quad f'(10)=20$

$\Rightarrow f(10)+f'(10)=120$

Illustration 28

If $f:R\to R$ satisfies $f(x+y)=f(x)+f(y)$, for all $x,y\in R$ and $f(1)=2$, then $\sum\limits_{r=1}^{7}f(r)$ is:

Ans. (56)

Solution:

$f(x+y)=f(x)+f(y)$

function should be $y=mx$

$f(1)=2\ \therefore m=2\ \Rightarrow\quad f(x)=7x$

$\sum\limits_{r=1}^{n}f(r)=2\sum\limits_{r=1}^{n}r=\dfrac{2n(n+1)}{2}$.

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Illustration 29

If $f'(2) = 4$ then, evaluate $\lim\limits_{x\to 0} \dfrac{f(1+\cos x)-f(2)}{\tan^2 x}$.

Ans. (–2)

Solution:

Given $f'(2) = 4$

$\lim\limits_{x\to 0} \dfrac{f(1+\cos x)-f(2)}{\tan^2 x}$ $\left[\dfrac{0}{0} \text{ form}\right]$ $\left[\dfrac{0}{0} : \text{i}\right]$

apply L'Hospital rule

$$\lim_{x\to 0} \frac{-\sin x \ f'(1+\cos x)}{2\tan x \sec^2 x} = \lim_{x\to 0} \frac{-\left(\frac{\sin x}{x}\right) f'(1+\cos x)}{2\left(\frac{\tan x}{x}\right)\sec^2 x} = -2$$

Illustration 30

Let a function $f : R \to R$ be given by $f(x + y) = f(x)\, f(y)$ for all $x, y \in R$ and $f(x) \neq 0$ for any $x \in R$. If the function $f(x)$ is differentiable at $x = 0$, show that $f'(x) = f'(0)\, f(x)$ for all $x \in R$. Also, determine $f(x)$.

Ans. ($f(x) = e^{xf'(0)}$ $\forall\ x \in R$)

Solution:

$f : R \to R$ and $f(x + y) = f(x)\, f(y)$ $\forall\ x, y \in R$

Put $x = y = 0$ $f(0) = f^2(0)$ since $f(0) \neq 0$

$f(0) = 1$

$$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = \lim_{h\to 0} f(x)\ \frac{[f(h)-1]}{h}$$

$= f(x)\, f'(0)$

Let $f(x) = y \Rightarrow \dfrac{dy}{dx} = y.f'(0)$

On solving $\ell ny = x\, f'(0) + c$

$y = f(x) = e^c . e^{xf'(0)}$

$\because f(0) = 1 \Rightarrow c = 0$

Thus $f(x) = e^{x.f'(0)}$ $\forall\ x \in R$

Illustration 31

Let $f(x)$ be a polynomial function satisfying the relation $f(x).f\left(\dfrac{1}{x}\right) = f(x) + f\left(\dfrac{1}{x}\right)$ $\forall\ x \in R - \{0\}$ and $f(3) = -26$. Determine $f'(1)$.

Ans. (–3)

Solution:

$f(x).f\left(\dfrac{1}{x}\right) = f(x) + f\left(\dfrac{1}{x}\right) \Rightarrow f(x) = 1 \pm x^n$

$f(3) = -26 \Rightarrow f(x) = 1 - x^3$

$f'(x) = -3x^2$ or $f'(1) = -3$

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Illustration 32

Let function $f(x)$ satisfying the relation $f(x+y)+f(x-y)=2f(x).f(y)$, then prove that it is even function

Solution:

Given $f(x+y)+f(x-y)=2f(x).f(y)$ ...(1)

Putting $y=x$ in equation (1), we get

$f(2x)+f(0)=2(f(x))^2$ ...(2)

Putting $y=-x$ in equation (1), we get

$f(0)+f(2x)=2f(x).f(-x)$ ...(3)

(3) – (2) $\Rightarrow 2(f(x))^2=2f(x)f(-x)$ $\Rightarrow f(x)=0$ or $f(x)=f(-x)$

in both cases, function is an even function

Illustration 33

Let $f(x)$ be a bounded function. $L_1=\lim\limits_{x\to\infty}(f'(x)-\lambda f(x))$ and $L_2=\lim\limits_{x\to\infty}f(x)$ where $\lambda>0$. If L_1, L_2 both exist and $L_1=L$, then prove that $L_2=-\frac{L}{\lambda}$.

Solution:

$$L_1=\lim_{x\to\infty}(f'(x)-\lambda f(x)) \quad\Rightarrow\quad L_1=\lim_{x\to\infty}\frac{e^{-\lambda x}\{f'(x)-\lambda f(x)\}}{e^{-\lambda x}}$$

[As it is a bounded function]

$$\Rightarrow\quad L_1=\lim_{x\to\infty}\frac{e^{-\lambda x}f'(x)-\lambda f(x)e^{-\lambda x}}{e^{-\lambda x}} \quad\Rightarrow\quad L=L_1=\lim_{x\to\infty}\frac{\frac{d}{dx}\{f(x)e^{-\lambda x}\}}{\frac{d}{dx}\left(-e^{-\lambda x}.\frac{1}{\lambda}\right)}$$

$$\Rightarrow\quad L=\lim_{x\to\infty}\frac{f(x)e^{-\lambda x}}{\left(-\frac{e^{-\lambda x}}{\lambda}\right)} \quad\Rightarrow\quad L=\lim_{x\to\infty}-\lambda f(x) \quad\Rightarrow\quad L=-\lambda\lim_{x\to\infty}f(x)$$

$$\Rightarrow\quad L=-\lambda L_2 \quad\Rightarrow\quad L_2=-\frac{L}{\lambda}$$

Illustration 34

Let R be the set of real numbers and $f: R\to R$ be such that for all x & y in R $|f(x)-f(y)|\le|x-y|^3$. Prove that $f(x)$ is constant.

Solution:

$f: R\to R$ $\quad |f(x)-f(y)|\le|x-y|^3$ $\quad \forall\, x, y\in R$

$$\left|\frac{f(x)-f(x)}{x-y}\right|\le|x-y|^2$$

taking lim $x\to y$ on both side we have

$|f'(y)|\le 0$ $\Rightarrow$ $f'(y)=0$ $\Rightarrow$ $f(y)$ is constant $\forall\, y\in R$

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Illustration 35

Let $f:\mathbb{R}\to\mathbb{R}$ be a differentiable function that satisfies the relation $f(x+y)=f(x)+f(y)-1,\ \forall x,\ y\in\mathbb{R}$. If $f'(0)=2$, then $|f(-2)|$ is equal to ____.

Ans. (3)

Solution:

$f(x+y)=f(x)+f(y)-1$

$$f'(x)=\lim_{h\to0}\frac{f(x+h)-f(x)}{h}$$

$$f'(x)=\lim_{h\to0}\frac{f(h)-f(0)}{h}=f'(0)=2$$

$$f'(x)=2\Rightarrow dy=2dx$$

$y=2x+C$

$x=0,\ y=1,\ c=1$

$y=2x+1$

$$|f(-2)|=|-4+1|=|-3|=3$$

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